Foundations of the Geometric Vacuum Framework: A Synthesis of the Spectral Graph Theory Arc*

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The Geometric Vacuum (GeoVac) framework is a discretization scheme for non-relativistic quantum mechanics on natural geometries, built on a packed lattice of quantum-number labels. For hydrogen the lattice's nodes are exactly the labels of the unit-S^3 harmonics, on which (minus) the continuous Laplace–Beltrami operator acts as the diagonal $SO(4)$ Casimir at every cutoff; the lattice's edge Laplacian does *not* replace or converge to that operator (it splits into $n_{\max}$ disconnected l-blocks). This synthesis paper assembles the framework's foundational arc—the packing construction, the discrete graph and its dynamics, the conformal equivalence with the unit S^3, the exchange constant taxonomy, the angular sparsity theorem, the Bargmann–Segal lattice, and the universal/Coulomb partition, together with the *reconvergence* node where these threads cohere (the tensor-product triple's two-body interaction and where it stops, the period census, the Tannakian substrate and cosmic-Galois comparison homomorphism, and the forced/free seam)—into a single coherent narrative. The arc has a dependency structure worth naming: a single taproot (the packing construction, made physical by the Fock S^3 projection) fans out into the framework's machinery, then reconverges at the periods, Tannakian, and forced/free nodes, where the scattered structural findings cohere into one statement about what the framework forces and what it consumes. Four headline structural results frame the program: [**SYMBOLIC PROOF**] (a) the conformal chain *from the continuum operator onward* — the unit-S^3 Laplace–Beltrami operator, its integer spectrum ([**PRIOR ART**], checked for $n \leq 4$), and the energy-shell projection, under which the hydrogen bound-state Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ — audited by a suite of 18 symbolic checks, with the non-definitional energy step (the S^3 Coulomb kernel's eigenvalue $2\pi^2/n$ forcing $p_0 = Z/n$) carried by the exact kernel-eigenvalue test of Paper 7 ($n \leq 4$; general n is Fock's); [**WALL**] the discrete Fock graph's edge Laplacian does *not* converge to that operator (it splits into $n_{\max}$ disconnected l-blocks with an $n_{\max}$-dimensional kernel); what is exact is its label set, on which the operator is the diagonal $SO(4)$ Casimir (the 18 checks do not touch the graph-side step) (ten of the eighteen limiting-case, definitional or non-discriminating rather than logically independent, Paper 7); [**PRIOR ART mechanism; the explicit densities are new**] (b) the angular sparsity of two-body matrix elements depends only on $l_{\max}$ and is bit-identical across five distinct central potentials, qualifying as a potential-independent theorem about spherical-fermion bases; [**PANEL-VERIFIED**] (c) the three-dimensional isotropic harmonic oscillator admits a discrete graph encoding on the Hardy space $H^2(S^5)$ that is bit-exactly π-free in exact rational arithmetic at every finite cutoff; [**OBSERVATION**] (d) the framework's results split cleanly along an A/D partition—an algebra-only universal sector that transfers free across central potentials, and a Dirac-operator-specific sector containing the spectrum, the projection, and the candidate fine-structure-constant combination of Paper 2 [22]. [**OBSERVATION**] A structural reading we make explicit in this synthesis is that the framework's substrate-level scale-freedom—the packing diameter d_0 cancelling out of every spectral quantity, and physical scales entering only through the Fock-projection focal length p_0—is the geometric expression of the same principle that conformal field theory embodies at the physics level (cf. the F-theorem of [31]). GeoVac provides a scale-free mathematical substrate; CFT is a scale-free physics theory living on such a substrate. This is not a claim that GeoVac is a CFT; it is a structural reflection consistent with (not evidenced by) Paper 50's closed-form re-derivation of the continuum CFT-on-sphere free energies, in which the truncation never enters (operator-algebra arc, outside the foundations scope of this synthesis). We present GeoVac as a discretization framework that exploits natural geometry, not as a new foundation for quantum mechanics: the discrete label lattice and the continuous Laplace–Beltrami operator are dual descriptions linked by an exact Peter–Weyl label identification, with Fock's conformal projection linking the operator to the Schrödinger equation (the bound-state Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$), and the interpretive question of which description is more fundamental is left open. The framework's concrete computational advantages—$O(V)$ sparsity, angular momentum selection rules baked into the basis, bit-exactly rational squared matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound—are independent of that interpretive question. (The integer spectrum $n^2 - 1$ is a property of the continuum operator, not of the graph, whose closed-form spectrum is bounded in $[0, 8]$.)

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* Group 3 synthesis in the Geometric Vacuum series

I. INTRODUCTION

The standard formulation of non-relativistic quantum mechanics presupposes a continuous configuration space and treats the Schrödinger operator $-\frac{1}{2}\nabla^2 + V(r)$ as the fundamental object. For the hydrogen atom in particular, Fock showed in 1935 [1] that the momentum-space Schrödinger equation maps by stereographic projection onto the unit three-sphere S^3: the bound states become the hyperspherical harmonics (the eigenfunctions of the Laplace–Beltrami operator Δ_{S^3}), and the bound-state Hamiltonian becomes the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of that operator. This observation has been developed extensively by Bargmann [2], Bander and Itzykson [3], Avery [10], Aquilanti and co-workers [11], and others working in the hyperspherical-momentum-space tradition, which uses it both for analytic work and as a computational method (Sturmian and hyperspherical-harmonic bases for atoms, molecules and reactive scattering). In that tradition the S^3 structure is a reformulation of a known continuum problem, not a foundation in its own right.

The Geometric Vacuum (GeoVac) framework [19, 20, 24] takes a complementary computational stance. Starting from a packing construction on an isotropic sphere, it builds a finite directed graph whose nodes are quantum-number labels (n, l, m) and whose edges are transition amplitudes generated by the $SU(2) \otimes SU(1,1)$ ladder algebra of hydrogen. The label set is dimensionless and scale-invariant; physical energies enter only through Fock's energy-shell constraint $p_0^2 = -2E$, which acts as a stereographic focal length, with $p_0 = Z/n$ forced by the eigenvalue of Fock's S^3 Coulomb kernel on the shell labelled n (not through the graph Laplacian, whose spectrum is a grid band structure). **[WALL]** This discrete construction does not converge to the unit-S^3 Laplace–Beltrami operator: the graph Laplacian has no l-changing edges, so it splits into $n_{\max}$ disconnected blocks, its saturation $\lambda_{\max} \to 8$ is the band edge of two-dimensional grids, and the s/p proxy decay [retracted 2026-09-04: sp-splitting-as-convergence-evidence] tracks cutoff divisibility ($\lambda_{2s} = 3$ exactly iff $n_{\max} \equiv 0 \bmod 3$; Paper 1 §III). What is exact is the label set: "degeneracy recovery" legitimately means that the nodes, n^2 per shell, are the Gel'fand–Tsetlin labels of the Peter–Weyl truncation of $L^2(S^3)$, on which $-\Delta_{S^3}$ is the diagonal $SO(4)$ Casimir $n^2 - 1$. The discrete labels and the continuous operator are linked by this exact Peter–Weyl label identification, with Fock's conformal projection linking the operator to the Schrödinger equation ($H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$), and we present them throughout as dual descriptions of the same physics; the interpretive question of ontological priority is explicitly left to the reader.

GeoVac differs from three nearby families of discretization schemes in ways worth naming at the outset. Lattice gauge theory discretizes the spacetime continuum on a hypercubic grid; GeoVac discretizes the quantum-number space and inherits the natural geometry of separable quantum systems. Tensor-network methods choose a basis adapted to entanglement structure; GeoVac chooses a basis adapted to angular momentum and the Fock conformal projection. Algebraic quantum field theory works at the level of operator algebras over Minkowski spacetime; GeoVac is non-relativistic and operates on compact Euclidean-signature manifolds. The rhetoric rule that governs the GeoVac papers (see the project's positioning statement) applies here: GeoVac is not a new foundation for quantum mechanics, and the synthesis presented in this paper does not assert ontological priority of either the discrete or the continuum description.

The Group 3 (foundations) arc comprises twelve papers, organized by a single dependency structure. *The taproot:* Paper 0 [19] states the packing axiom and, with structural choices motivated rather than forced, reads the angular quantum numbers as packing indices; Paper 7 [24] makes it physical, verifying the conformal equivalence with the unit S^3 via a suite of 18 symbolic checks (ten of the eighteen limiting-case, definitional or non-discriminating) plus an exact test (for $n \leq 4$; general n is Fock 1935, a Funk–Hecke instance) that the S^3 Coulomb kernel's eigenvalue $2\pi^2/n$ forces the focal length $p_0 = Z/n$, and extending to the multi-electron S^{3N-1} setting. *The fan-out* (the framework's working machinery): Paper 1 [20] sets up the spectral graph theory and reproduces the Rydberg quantum numbers on interior shells (the energies are not a graph output: in Paper 7 the Coulomb kernel's eigenvalue forces $p_0 = Z/n$ and the energy shell then gives E_n); Paper 18 [25] tags transcendental content via an exchange-constant taxonomy and the master Mellin engine (a census and a reading); Paper 22 [27] states the potential-independent angular sparsity theorem; Paper 24 [28] constructs the Bargmann–Segal lattice and the Coulomb/HO asymmetry (seven layers catalogued, five standing); Paper 31 [29] states the $\mathcal{A}/D$ universal/Coulomb partition in spectral-triple language. *The reconvergence* (where the branches loop back to a unified foundational statement, §IX): Paper 54 [40] gauges the tensor-product triple into a two-body interaction consistent with the selection rules but only their monopole, direct subset, Paper 55 [41] places the transcendentals in the census of GeoVac spectral data in cyclotomic mixed-Tate periods at level ≤ 4, Paper 56 [42] constructs a depth-1 cosmic-Galois comparison, a homomorphism from the level-4 unipotent radical $\mathcal{U}_4$ into GeoVac's abelian factor, through $\mathcal{U}_4^{\mathrm{ab}}$, and Paper 57 [43] catalogues the forced/free seam; Paper 61 [45] develops the arithmetic of the elliptic layer (§IX). An $O(V)$ time-evolution engine (Paper 6 [23], now an archived standalone tool) supplies the dynamical demonstration—an integrator check run on the labelled legacy graph model, not on physical hydrogen (§V); this synthesis supersedes the earlier single-document synthesis (Paper 21 [26]).

Four headline structural results organize the rest of this synthesis. **(H1) [SYMBOLIC PROOF]** The con-

formal chain from the unit-S^3 Laplace–Beltrami operator to the Schrödinger equation is audited symbolically (Paper 7 [24]): Fock's projection makes the bound-state Hamiltonian the function $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ of that operator, its scale $p_0 = Z/n$ fixed by the kernel eigenvalue $2\pi^2/n$ (tested exactly for $n \leq 4$; general n is Fock's); **[WALL]** the discrete Fock graph's edge Laplacian does not converge to that operator: the proved spectral-bound statement — the saturation of the bipartite bound, **[INTERNAL THEOREM]** closed form with rate $C_{\mathrm{sat}}/n_{\max}^2$ (Paper 0 §VI) — is a grid band edge, and the size of the s/p proxy tracks the residue of $n_{\max}$ modulo 3, not convergence (the label set is exact), while **[INTERNAL THEOREM]** the state-space Gromov–Hausdorff convergence of the spinor spectral truncations (the commutative S^3 factor, true window; the almost-commutative extension **[OPEN]**) is proved unconditionally: the distance is bounded by a closed-form Fejér moment $\gamma_{n_{\max}}$ that vanishes in the limit. For the Fejér kernel, $\gamma_{n_{\max}} = (4/\pi)\log n_{\max}/n_{\max} + O(1/n_{\max})$, with the constant $4/\pi$ (dual-Coxeter normalisation; $2/\pi$ on the unit S^3, $2\sqrt{2}/\pi$ under the field-standard normalised form $(\theta|\theta) = 2$ of Kac [21], where $\mathrm{Cas(ad)} = 2h^\vee$ (computed for $\mathfrak{su}(2), \mathfrak{su}(3)$ in `tests/test_kac_normalisation_casimir.py`)) tiered **[INTERNAL THEOREM + MEASURED]**; the constant is the kernel's, not the convergence rate, since a tapered amplitude gives a **[MEASURED]** $O(1/n_{\max})$ (Paper 38 [37], main theorem). **(H2) [PRIOR ART mechanism; explicit densities new]** The angular sparsity density of two-body matrix elements is potential-independent (Paper 22 [27]): a structural theorem of spherical-fermion bases, not a Coulomb-specific accident. **(H3) [PANEL-VERIFIED]** The 3D harmonic oscillator admits a discrete graph encoding on the Hardy sector of S^5 that is bit-exactly π-free (Paper 24 [28]). **(H4) [OBSERVATION]** The framework's results decompose cleanly into a universal algebra-only sector and a potential-specific Dirac-dependent sector (Paper 31 [29]). We discuss the fine-structure constant combination $K = \pi(B + F - \Delta)$ of Paper 2 [22] only as an Observation, in keeping with the project rule that the combination rule remains an *Observation* at the rule level despite three spectral homes of distinct type for B, F, and Δ.

II. THE PACKING CONSTRUCTION AND THE GRAPH

GeoVac's discrete lattice is constructed by a packing argument that is prior to the Schrödinger equation [19]. Two axioms suffice, together with the construction's structural choices (motivated, not forced).

Axiom 1 (binary distinguishability). A minimum of two distinguishable states is required to define a metric on an information-bearing manifold.

Axiom 2 (geometric indifference). In the absence of external constraints, the distribution of states on a homogeneous surface maximizes entropy and is therefore isotropic.

[SYMBOLIC PROOF] Given concentric isotropic shells in two dimensions with linearly spaced radii and area-proportional state counting (choices the axioms motivate but do not force, Paper 0), the angular state count on shell k is $2k - 1$. The shell capacities $\{1, 3, 5, 7, \ldots\}$ equal in count the spherical harmonic basis Y_{lm} under the identification $l = k - 1$ and $m = -l, \ldots, +l$ (attaching the labels to positions is a labelling choice). The per-shell degeneracy $2l + 1$ is thus reproduced as an exact count from the packing geometry and these counting choices, without invoking the Schrödinger equation or the angular momentum operators $\hat{L}^2$, $\hat{L}_z$.

[OBSERVATION] The principal quantum number n enters as a cumulative grouping of the first n shells, producing the count $\sum_{k=1}^{n}(2k-1) = n^2$. Paper 0 is careful to flag that this grouping is natural but not uniquely forced by the two axioms. Similarly, the factor of two in the per-shell count, set by the binary initialization (with the orientability of the compactified packing manifold S^2 as a heuristic reading), corresponds in counting to spin multiplicity, which the paper presents as a structural correspondence rather than a derivation. The strong claim is the angular sector (l, m); the other identifications are framed as structural correspondences.

From the packing to the graph

The packed lattice acquires a graph structure when the $SU(2) \otimes SU(1,1)$ ladder operators of the hydrogen dynamical algebra [4, 18] are imposed as edges:

$$L_\pm|n, l, m\rangle \propto |n, l, m \pm 1\rangle, \tag{1}$$

$$T_\pm|n, l, m\rangle \propto |n \pm 1, l, m\rangle. \tag{2}$$

The angular operators $L_\pm$ connect states within an n-shell along m; the radial operators $T_\pm$ connect states between adjacent n-shells at fixed l. **[SYMBOLIC PROOF]** Paper 1 [20] shows that the operators built from the ladder generators reproduce the Rydberg quantum numbers exactly on interior shells: the $SU(1,1)$ element $N = -2[T_+, T_-]$ returns n for $n < n_{\max}$ (not on the truncated top shell), and the $SU(2)$ Casimir returns $l(l+1)$; on every shell the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ is exact on the labels. The energies $E_n = -1/(2n^2)$ are not an output of the ladder operators or of the graph (Paper 1 imposes them as a map from n); the Casimir yields them only through the function $E_n = -Z^2/(2(C+1))$, $C = n^2 - 1$, that Fock's kernel fixes (Paper 7); in Paper 7's continuum chain the eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel forces $p_0 = Z/n$, and the energy shell $p_0^2 = -2E$ then gives E_n. At the algebraic level the discrete and continuous descriptions therefore agree on the quantum-number skeleton, while the graph Laplacian's node-amplitude proxy for the s/p gap is large at finite

cutoff (up to 37% at $n_{\max} = 5$ on the binary lattice; 0.39% at $n_{\max} = 30$, which sits on the $n_{\max} \equiv 0 \bmod 3$ branch, against 1.65% and 2.58% at 28 and 29; the ℓ-blocks are disconnected, so this is a proxy and not a spectral gap — Paper 1 §III).

The kinetic constant $\kappa = -1/16$

[OBSERVATION] A second constant attached to the packing graph is the kinetic constant $\kappa = -1/16$. Its role is narrow: it matches the graph's extremal eigenvalue (a band edge, $\lambda_{\max} \to 8$) to the hydrogen ground state $-\frac{1}{2}$. It is not the energy unit of the Rydberg ladder and does not map the graph spectrum onto it: the Rydberg levels come from Fock's continuum projection, with no role for κ: the eigenvalue $2\pi^2/n$ of the S^3 Coulomb kernel on the shell labelled by n forces the focal length $p_0 = Z/n$, and the energy shell $p_0^2 = -2E$ then gives E_n (Paper 7; the Casimir $C = n^2 - 1$ alone does not fix the function, but once Fock's kernel fixes it, $E_n = -Z^2/(2(C+1))$ on the labels). Recent analysis [24, 25] has clarified its status: the value $\kappa = -1/16$ *coincides* with a geometric quantity computed independently of any energy calibration. The squared coupling between adjacent n-shells in the Gegenbauer eigenbasis is $c^2(n,l) = (1/4)[1 - l(l+1)/(n(n+1))]$ (corrected 2026-09-03 from a printed 1/16 prefactor; the Chebyshev amplitude is 1/2), giving $c^2(n,0) = 1/4$ for $l = 0$; the quantity that coincides with 1/16 is the fourth power of the inverse conformal factor, $1/\Omega^4(0)$, equivalently $c^2(n,0)/4$. That the energy-calibration constant κ *equals* this geometric 1/16 is recorded as an *Observation*, not a derivation: the geometric value is rigid, but no mechanism is known that derives κ's calibration role from it (Paper 7 §III). Paper 18 accordingly lists κ beside the *conformal* tier's energy shell p_0 only as a matched scale: it is the energy shell, not κ, that carries the spectrum into atomic units.

Status of the packing-versus-physics question

[OBSERVATION] The discrete lattice is built by the packing construction, before any reference to the Coulomb potential or the Schrödinger equation, though its radius and counting choices are motivated rather than forced (Paper 0). It matches known atomic structure—the Rydberg quantum-number labels and the chemical periodicity of multi-electron atoms (Paper 16 [33], outside the foundations group). We present this as the *structural fit* between the packing construction and known atomic structure. The synthesis paper does not argue for the ontological priority of either description; the dual-description framing of Paper 7 governs the discussion.

III. THE S^3 CONFORMAL EQUIVALENCE

The theoretical foundation of GeoVac is the conformal equivalence, established in Paper 7 [24], between the *continuum* unit-S^3 Laplace–Beltrami operator and the hydrogen Schrödinger problem, carried by the Fock projection (the bound-state Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$). The discrete Fock graph's role is its label set, on which minus that operator is the diagonal $SO(4)$ Casimir exactly; that the graph's edge Laplacian has that operator as its continuum limit is *false* (it splits into $n_{\max}$ disconnected l-blocks). This section states the result and explains its operational role.

The Fock projection and the chordal distance identity

[SYMBOLIC PROOF] Fock's 1935 construction [1] writes the momentum-space hydrogenic (nuclear charge Z) Schrödinger equation

$$\left[\frac{p^2}{2} - E\right]\varphi(\mathbf{p}) - \frac{Z}{2\pi^2}\int \frac{\varphi(\mathbf{p}')}{|\mathbf{p}-\mathbf{p}'|^2}\,\mathrm{d}^3p' = 0 \qquad (3)$$

and applies the stereographic projection

$$u_i = \frac{2p_0\,p_i}{p^2+p_0^2}, \quad u_4 = \frac{p^2-p_0^2}{p^2+p_0^2}, \quad p_0 = \sqrt{-2E}, \qquad (4)$$

which maps $\mathbf{p} \in \mathbb{R}^3$ onto the unit three-sphere $\mathbf{u} \in S^3 \subset \mathbb{R}^4$. The Coulomb kernel $1/|\mathbf{p}-\mathbf{p}'|^2$ becomes the inverse-square chordal distance on S^3, and with $\varphi = \Omega^2\Psi$ the integral equation reduces to the S^3 integral equation

$$p_0\,\Psi(\boldsymbol{\omega}) = \frac{Z}{2\pi^2}\int_{S^3}\frac{\Psi(\boldsymbol{\omega}')}{|\boldsymbol{\omega}-\boldsymbol{\omega}'|^2}\,\mathrm{d}\Omega' \qquad (5)$$

(Paper 7). The kernel is zonal, so it is diagonal on the S^3 harmonics; on the degree-$(n-1)$ harmonics its eigenvalue is $2\pi^2/n$ (Fock 1935, an instance of Funk–Hecke; tested exactly for $n \le 4$ in Paper 7), and this is what forces the focal length $p_0 = Z/n$ and hence $E_n = -Z^2/(2n^2)$. The Laplace–Beltrami operator carries the same content in operator form: the degree-$(n-1)$ harmonics are the eigenfunctions of $-\Delta_{S^3}$ with eigenvalue $C = n^2-1$ (the $SO(4)$ Casimir; an integer spectrum fixed by the round unit-sphere geometry, the metric and not merely the topology, Paper 7), so the kernel acts as $2\pi^2(1-\Delta_{S^3})^{-1/2}$ and the bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$, $E_n = -Z^2/(2(C+1))$. This is the precise sense in which Fock's projection makes Δ_{S^3} equivalent to the Schrödinger problem (the standard "free motion on S^3" reading). The kernel is not the Green's function of Δ_{S^3} (that behaves like $1/d$, with eigenvalues $\propto 1/(n^2-1)$); it is the restriction to S^3 of the Green's function of the $\mathbb{R}^4$ Laplacian (Paper 7).

The 18 symbolic checks

[**SYMBOLIC PROOF**] Paper 7 [24] verifies the conformal-equivalence chain through a suite of 18 symbolic checks implemented in `sympy` (Paper 7 flags ten of them as limiting-case, definitional or non-discriminating rather than logically independent theorems). The checks cover the stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal factor identity, the conformal Laplacian transformation law, the integer eigenvalue spectrum, and the (definitional) eigenvalue-to-energy bookkeeping of the energy-shell projection from S^3 to $\mathbb{R}^3$. These tests live in `tests/test_fock_projection.py` and `tests/test_fock_laplacian.py` in the GeoVac repository and are run as topological integrity checks before any release. The non-definitional part of the energy step is carried separately by `tests/test_paper7_fock_kernel.py`: in exact arithmetic, the eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel on the degree-$(n-1)$ harmonics (every $l < n$ for $n \leq 4$, one real harmonic per l; [**PRIOR ART**] the eigenvalue holds for every n, Fock 1935 / Funk–Hecke), the resulting $p_0 = Z/n$ with E_n never an input, and the identification of the Fock harmonics with the hydrogen momentum wavefunctions. Their status as machine-verified algebraic identities pins the continuum-side conformal chain; the discrete-to-continuum step itself lies outside their scope (Paper 7 states this explicitly).

Schrödinger recovery: labels, not a continuum limit of L

[**WALL**] The graph Laplacian on the GeoVac lattice does *not* converge, as $n_{\max} \to \infty$ under any rescaling, to the round S^3 Laplace–Beltrami operator: its edges never change l, so it is the direct sum of $n_{\max}$ disconnected grid graphs, with $\dim\ker = n_{\max}$ at every cutoff where the S^3 Laplacian has one zero mode — the load-bearing obstruction — for the binary production weights and the $1/(n_i n_j)$ option alike (the graph heat kernel also vanishes between blocks, but reading that as a continuum statement would need a node-to-point identification the papers do not make) (`tests/test_graph_s3_nonconvergence.py`). The closed-form saturation of the spectral bound, [**INTERNAL THEOREM**] (Paper 0 §VI), is a grid band edge, and the size of the s/p proxy tracks the residue of $n_{\max}$ modulo 3, not convergence. What is exact, at every cutoff, is the label set: the nodes are the Gel'fand–Tsetlin labels of the Peter–Weyl truncation of $L^2(S^3)$, and on them $-\Delta_{S^3}$ is the diagonal $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ (the Casimir value is checked from the generators in the same test; its identity with $-\Delta_{S^3}$ is prior art, not tested there). What is a theorem on the operator-algebra side is a statement about a different object: [**INTERNAL THEOREM**] for the spinor (Dirac) spectral truncations (the commutative S^3 factor, true window; the almost-commutative extension [**OPEN**]) it is an unconditional van Suijlekom state-space Gromov–Hausdorff convergence, bounded by a closed-form Fejér moment $\gamma_{n_{\max}} \to 0$ (proved), under the translation-seminorm metrization (Paper 38 [37], operator-algebra arc); the Fejér moment behaves as $(4/\pi)\log n_{\max}/n_{\max}$, its constant $4/\pi$ (dual-Coxeter normalisation; $2/\pi$ on the unit S^3, $2\sqrt{2}/\pi$ under the field-standard normalised form $(\theta|\theta) = 2$ of Kac [21], where $\mathrm{Cas}(\mathrm{ad}) = 2h^\vee$ (computed for $\mathfrak{su}(2), \mathfrak{su}(3)$ in `tests/test_kac_normalisation_casimir.py`)) tiered [**INTERNAL THEOREM + MEASURED**] (derived and numerics-pinned; a kernel constant, not the convergence rate: a tapered amplitude gives a [**MEASURED**] $O(1/n_{\max})$), with the extension to compact connected semisimple Lie groups (Paper 40 [38]: the convergence conditional on the per-group spin-window decomposition; whether the Fejér-moment constant is the same for every simple group is open); the scalar analogue of that theorem, for the Peter–Weyl spectral truncations of $L^2(S^3)$, is the prior-art state-space GH result of Gaudillot-Estrada–van Suijlekom [46]. Paper 7's 18 symbolic checks (ten of the eighteen limiting-case or definitional) target the conformal-transformation chain, not a convergence statement, and the graph-side *operator* convergence is refuted. The saturation of the bipartite bound $\lambda_{\max} \to 2d_{\max} = 8$ *is* a theorem — [**INTERNAL THEOREM**] closed form with rate $C_{\mathrm{sat}}/n_{\max}^2$, $C_{\mathrm{sat}} = 42.7397\ldots$ (Paper 0 §VI; `tests/test_paper1_block_spectrum.py`), a Dirichlet band edge of the grid blocks — and the frozen test `tests/test_paper7_graph_convergence.py` pins it together with the block structure of the lattice Laplacian; neither is operator convergence. The eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel on the shell with $\lambda_n = -(n^2-1)$ forces the focal length $p_0 = Z/n$ (Paper 7, exact test for $n \leq 4$; Fock 1935 for all n), and the energy-shell constraint $p_0^2 = -2E$ then gives the hydrogenic Rydberg series $E_n = -Z^2/(2n^2)$; the Casimir $C = n^2 - 1$ alone does not fix that function, but once Fock's kernel fixes it, $E_n = -Z^2/(2(C+1))$ on the labels, i.e. $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$. The dimensionless label set carries no scale; the Rydberg energy is injected exclusively at the projection step. This separation between dimensionless topology and dimensional projection is the *dimensionless vacuum* viewpoint of Paper 7 (the scale invariance itself is Fock's, explicit in Bander–Itzykson 1966 [3], [**PRIOR ART**]): all physical dimensions enter exclusively through the energy-shell constraint, i.e. the focal-length parameter p_0 (the calibration π content enters separately, read as the $k = 0$ volume slot of §VII, a reading with a [**CONDITIONAL**] engine).

Scale-freedom of the substrate and its reflection in conformal field theory

[**OBSERVATION**] The separation between dimensionless topology and dimensional projection described above is more than a computational convenience. It is the geometric expression of a principle that physics encounters in another guise: conformal field theory.

A conformal field theory is a quantum field theory whose action and correlators are invariant under rescaling of the underlying coordinates, together with the larger conformal group $SO(d, 2)$ in d spacetime dimensions. Operationally: a CFT has no preferred length scale, and its observables transform predictably under conformal transformations. The standard examples include free massless scalars and fermions on round S^d, where the partition functions $Z_{S^d}^{\mathrm{CFT}}$ are computable in closed form and reproduce universal coefficients (e.g. the F-theorem constant F_{S^3} [31]).

GeoVac's substrate exhibits an analogous property at the *geometric / mathematical* layer, prior to physics. The packing construction (Paper 0 [19]) takes a circle of diameter d_0 (the fundamental scale, with no specific physical value assigned) and outputs node counts equal to the dimensions of the eigenspaces of the continuum S^3 operator (attaching the labels (ℓ, m) to the nodes is a labelling choice, Paper 0) (the integer eigenvalues $n^2 - 1$ belong to that continuum operator, carried on the labels as the diagonal $SO(4)$ Casimir; the graph spectrum is not a level-by-level copy of it, Paper 7), the multiplicities are integers, the Wigner-symbol couplings are exact square roots of rationals (their squares rational), and the packing diameter d_0 *cancels out* of every spectral quantity. Shrink d_0 to the Planck length, grow it to a galactic radius, set it to anything in between: the substrate is the same. The labels that come out of the packing are atomic quantum numbers in interpretation, but they are not atomic-scale in construction. The packing produces a scale-free structural skeleton.

The Fock projection of Paper 7 [24] is a *conformal map*: stereographic projection from S^3 to $\mathbb{R}^3$ preserves angles but rescales lengths by a position-dependent conformal factor $\Omega(\mathbf{p}) = 2p_0/(p^2 + p_0^2)$. Physical scales enter the framework through exactly this factor, in the form of the focal-length parameter p_0. The scale-free substrate becomes scale-bearing physics only when a specific p_0 is supplied (e.g. $p_0 = Z/n$ for hydrogenic atomic levels). Different choices of p_0 correspond to different physical interpretations of the same combinatorial skeleton; the substrate is agnostic to the choice.

This is structurally the same principle that CFT scale-invariance embodies. A CFT is a physics theory with no preferred length scale at the level of its action and observables; GeoVac's label set, with the matrix elements built on it, is a mathematical object with no preferred length scale. CFT scale-invariance is the physics side of the principle; the dimensionless label set is the substrate side. The two converge structurally in that they realise the same property—scale-freedom—at different layers of description, and the Fock projection that connects them is the conformal map that implements the scale-free-to-scale-bearing transition.

This reading is not a claim that GeoVac *is* a CFT, nor that the discrete graph derives CFT scale-invariance. Those would overshoot: GeoVac provides a geometric / spectral substrate, and a CFT is a quantum field theory living on a substrate. Paper 50 (operator-algebra arc, outside the foundations scope of this synthesis) evaluates the standard zeta-regularised CFT-on-sphere free energies of the conformally coupled scalar and Camporesi–Higuchi Dirac on S^3 and S^5 in closed Hurwitz form from the full continuum spectra — a re-derivation of known continuum values in which the discrete truncation never enters, so it is not evidence from the discrete substrate. It is merely consistent with the substrate-side scale-freedom established here; no mechanism is claimed. The substrate is the right kind of mathematical object to host scale-free physics, which is a reading of the parallel, not an explanation of it.

Multi-electron extension to S^{3N-1}

Paper 7 [24] and the archived Paper 21 [26] extend the construction to N-electron atoms by replacing S^3 with the unit sphere $S^{3N-1} \subset \mathbb{R}^{3N}$ under the $SO(3N)$ rotation group (the label-set generalisation is **[PRIOR ART]**, standard hyperspherical-harmonic theory; Paper 7 retires the $N > 1$ graph-limit conjecture). The single-electron Fock projection becomes a multi-electron hyperspherical projection, and the N-fermion antisymmetry condition on S^{3N-1} reads as a representation-theoretic constraint on the angular sector. Paper 16 recasts the chemical periodicity of the periodic table in terms of the S_N subgroup of $SO(3N)$ acting on the hyperspherical angular basis, mapping onto, rather than predicting, the known table (see Paper 16, group 4, outside the foundations scope of this synthesis).

[PRIOR ART; SYMBOLIC PROOF] The classical closed-form single-center two-electron repulsion integral $F^0(1\mathrm{s}, 1\mathrm{s}) = 5Z/8$ is re-derived from the density-overlap formula of Paper 7 for s-orbital pairs — the standard Fourier–Parseval form of the Coulomb integral, written in the stereographic variable $t = q/(2Z)$ (the single-integral structure is Parseval's, not the S^3 geometry's). More generally, all single-center Slater F^k integrals reduce to exact rationals in arithmetic over $\mathbb{Q}$ (the hypergeometric Slater engine, `geovac/hypergeometric_slater.py`, which is not part of Paper 7's derivation). The single-center two-electron *matrix elements* are thus closed in rational arithmetic (exactness of the integrals, not accuracy of the truncated problem); transcendentals appear only at the projection-to-energy step (see Section VII).

What Paper 7 does and does not claim

Paper 7 is careful to separate established mathematics from new contribution. The stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal Laplacian relation, and the integer eigenvalue spectrum are all known results from Fock, Bargmann, Bander–Itzykson, and Barut–Kleinert. The new contributions are: (a) the proved spectral-bound saturation ([**INTERNAL THEOREM**], closed form, Paper 0 §VI) and the s/p-proxy decay [retracted 2026-09-04: sp-splitting-as-convergence-evidence] through $n_{\max} = 30$ — the latter now known to track the residue of $n_{\max}$ modulo 3 rather than convergence, and so withdrawn as evidence, 2026-09-03), once read as indicating that the discrete GeoVac graph converges to the continuous S^3 Laplace–Beltrami operator — a reading now withdrawn as a wall (the graph splits into $n_{\max}$ disconnected l-blocks; 2026-09-25), the exact content being the label set, on which minus that operator is the diagonal $SO(4)$ Casimir; (b) the systematic symbolic verification of the conformal transformation chain; (c) the organizing use of Fock's scale invariance, physical scales confined to the energy-shell projection (the property itself [**PRIOR ART**], Bander–Itzykson 1966 [3]); (d) the computational consequence that sparse Hamiltonian-matrix eigenvalue problems (adjacency nonzeros growing as $V^{1.07}$, measured) can replace continuous integration. (The label set itself is [**PRIOR ART**], standard $SO(4)$ representation theory; the new content of item (a) is the wall.) Paper 7's explicit statement that its results are "statements of mathematical equivalence, not claims about which formulation is physically prior" is the foundation of the dual-description framing carried through the rest of the foundations papers.

IV. ANGULAR SPARSITY AS A STRUCTURAL THEOREM

The angular sparsity theorem of Paper 22 [27] is the foundations-side statement of one of the two structural mechanisms (with the composed block decomposition) behind GeoVac's exactly Q-linear composed Pauli-term count ($N_{\mathrm{Pauli}} = 27.90 \times Q$ for main-group molecules at fixed basis, Paper 14 [32]) at the qubit-encoding level. The theorem can be stated cleanly without reference to the Coulomb potential or to any specific radial wavefunction.

Theorem (Paper 22). [PRIOR ART mechanism (Condon–Shortley, Slater); the explicit quantitative density at each $l_{\max}$ is the new content] *For any N-fermion system whose single-particle orbitals factor as products of spherical harmonics and arbitrary radial functions, the fraction of nonzero two-body matrix elements is determined solely by angular momentum selection rules and is independent of the radial potential.*

TABLE I. Angular ERI densities from Paper 22 (Theorem 3): the physically-correct global-M_L Coulomb density $D(l_{\max})$ ($m_a + m_b = m_c + m_d$), which *the composed pipeline realizes*, and the pair-diagonal density $D_{\mathrm{pd}}(l_{\max})$ (per-pair $m_a = m_c$, $m_b = m_d$), which holds only for axially-symmetric or m-decoupled bases and is shown for comparison. The "realizes" attribution was corrected 2026-08-30: the production enumerator returned D_{pd} only because of the wrong-sign-q coefficient bug fixed on 2026-08-29. Both are universal across Coulomb, harmonic oscillator, Woods–Saxon, square well, and Yukawa potentials.

$l_{\max}$	$D(l_{\max})$ (realized)	$D_{\mathrm{pd}}(l_{\max})$
0	100%	100%
1	14.84%	7.81%
2	8.52%	2.76%
3	6.06%	1.44%
4	4.83%	0.90%
5	3.99%	0.62%

The proof is a direct application of the Wigner–Eckart theorem and Condon–Shortley angular momentum algebra [15, 16]. The two-body matrix element of a rotationally invariant interaction factors into a sum of products of Gaunt coefficients (angular factor) and Slater F^k, G^k integrals (radial factor); the Gaunt coefficients vanish for quartets (l_a, l_b, l_c, l_d) that do not satisfy the $3j$-triangle inequality $|l_a - l_c| \le k \le l_a + l_c$ and the corresponding inequality for (l_b, l_d). Exact enumeration at fixed $l_{\max}$ yields the ERI densities reproduced in Table I.

Universality across potentials

[PANEL-VERIFIED] The bound has been verified numerically across five qualitatively distinct potentials—Coulomb $-Z/r$, isotropic harmonic oscillator $\frac{1}{2}m\omega^2 r^2$, Woods–Saxon nuclear single-particle potential, square well, and Yukawa $e^{-\mu r}/r$—and produces bit-identical sparsity patterns at matched orbital counts. The universality is structural: the radial-angular factorization of $1/r_{12}$ in spherical harmonics (used in quantum chemistry) and the Talmi–Moshinsky bracket factorization of central NN potentials in oscillator bases (used in nuclear shell models [17]) impose the same angular selection rules. GeoVac's contribution is to state the explicit quantitative density at each $l_{\max}$ and to frame it as a universal resource guarantee independent of the radial physics.

Spinor extension

[PANEL-VERIFIED] Paper 22's spinor extension shows that the angular sparsity bound survives, with an inflated prefactor (about 14× at matched $l_{\max}$, milder at matched qubit count), when the basis is lifted from Y_{lm} to the spinor spherical harmonics $\Omega_{\kappa m_j}$ labeled by the Dirac

quantum numbers (κ, m_j). The density $d_{\text{spinor}}(l_{\max})$ satisfies $d_{\text{spinor}} \leq d_{\text{scalar}}$ at every $l_{\max}$ tested, with the same triangle inequality controlling vanishing of the matrix elements at the spinor level. Tier-2 spin-orbit extensions of the framework therefore inherit the angular sparsity guarantee without modification.

Operational consequence

The angular sparsity theorem, together with the composed block decomposition, is what makes GeoVac's qubit-encoded Hamiltonians competitive for VQE/NISQ resource estimation. **[MEASURED]** The quantum-computing applications papers (Group 4) report Pauli-term counts exactly linear in the qubit count for the composed-geometry architecture across the molecular library ($N_{\text{Pauli}} = 27.90 \times Q$ for main-group molecules at fixed basis)—a 51× to 317× equal-qubit advantage in Pauli-term count over published Gaussian baselines (H_2O, $n_{\max} = 1$–3; the retired pair-diagonal rule gave $O(Q^{2.5})$ and 51×–1712×). The structural reasons for this advantage are the composed architecture's block decomposition (each electron group's integrals live in its own block, so the term count adds across blocks, which is what makes it linear in Q) together with the Gaunt-selection-rule sparsity inside each block, quantified by the realized density $D(l_{\max})$ of Table I; the foundations-paper statement is the theorem itself, not the specific molecular benchmarks.

V. QUANTUM DYNAMICS ON THE GRAPH

Paper 6 [23] extends the framework from spectroscopy to time evolution. *Provenance note:* Paper 6 is archived within the series. Of the results quoted in this section, the unitarity benchmark and the resonant Rabi oscillation (item (b) below) were re-run and reproduce, with regression backing live in `tests/test_rabi_oscillation.py`. Items (a), (c) and (d) below are Paper 6's archived measurements with no live regression test (a named coverage gap in the claim matrix). *Scope:* both live checks run on the labelled legacy graph model `AtomicSolver(hamiltonian='graph')`, $H = \kappa(D - A)$, whose excited spectrum is a grid band structure and not physical hydrogen (Paper 7); they are checks of the Crank–Nicolson integrator on that model, not hydrogen physics. The graph Hamiltonian H is sparse with $O(V)$ nonzero entries, so the Crank–Nicolson propagator

$$\psi(t + \Delta t) = \left(I + \tfrac{i}{2} H \Delta t\right)^{-1} \left(I - \tfrac{i}{2} H \Delta t\right) \psi(t) \qquad (6)$$

admits implementation at $O(V)$ cost per step using sparse matrix-vector products plus iterative linear solves. **[MEASURED]** Unitarity is preserved to 10^{-10} (measured 1.1×10^{-13}) over 10^3 time steps in the live graph-model regression ($n_{\max} = 10$, 385 states); Paper 6's original H_2 benchmark reported machine precision over 10^4 steps (archived measurement).

The applications demonstrated in Paper 6 are (tiers per item: only (b) is a live **[MEASURED]** result; (a), (c) and (d) are archived measurements, reported as historical record without live backing): (a) broadband molecular spectroscopy from a single delta-kick propagation on the superseded LCAO bridged-graph H_2 model (Paper 6), recovering 20 of that model's 35 dipole-active transitions with 0.16% mean error relative to the model's own direct diagonalization (not to experimental H_2), in 33 seconds; (b) resonant Rabi oscillations matching the analytic two-level period to 0.41% with 99.98% coherent population transfer (live run on the graph model, $n_{\max} = 4$, 30 states; the period is matched against the analytic formula evaluated at the model's own level gap, not hydrogen's, so this tests the integrator, not hydrogen spectroscopy); (c) molecular dynamics with nuclei coupled to graph forces via Velocity Verlet, conserving total energy to 0.0003% (archived Paper 6, on the superseded LCAO graph-concatenation construction whose potential surface has no equilibrium, FCI-M); (d) NVT-ensemble Langevin dynamics at 300 K on the same superseded LCAO construction (the "stable thermal vibrations" originally reported cannot be vibrations about an equilibrium that construction does not have, so this item is historical only). Only (b) and the unitarity benchmark carry live regression backing (see the provenance note above).

Paper 6 is not a foundational claim in the same sense as Papers 7, 18, 22, 24, or 31—it is an operational demonstration that the foundations support time evolution. Its role in the synthesis is to close the gap between the spectral statement of Papers 1 and 7 (the ladder operators $T_\pm, L_\pm$ reproduce the Rydberg quantum numbers on interior shells, Paper 1) and the dynamical statement required for any quantum-chemistry application (a sparse Hamiltonian on the labels supports time evolution at the same $O(V)$ cost; the live demonstration uses the graph model, not physical hydrogen). Paper 6's dynamics engine demonstrates that the framework's sparse Hamiltonians support sparse time propagation.

VI. THE BARGMANN–SEGAL LATTICE AND THE COULOMB/HO ASYMMETRY

Paper 24 [28] extends GeoVac to the three-dimensional isotropic harmonic oscillator (HO) and uncovers a structural asymmetry between the Coulomb and HO discretizations. The construction is built on the Bargmann–Segal transform [5, 6].

The discrete HO lattice on S^5

The Bargmann–Segal transform maps the 3D isotropic HO Hamiltonian to the Euler number operator $\hat{N}$ on the Fock–Bargmann space of holomorphic functions on $\mathbb{C}^3$. Restriction to the unit sphere $S^5 \subset \mathbb{C}^3$ identifies the eigenspaces with the symmetric $SU(3)$ irreducible representations $(N, 0)$, of dimension $(N+1)(N+2)/2$, which match the HO shell degeneracies exactly. The HO Hamiltonian becomes $\hbar\omega(\hat{N} + 3/2)$, with the eigenvalues linear in the shell index N.

The discrete graph encoding of this structure takes nodes (N, l, m_l) to be the labels of the $(N, 0)$-restricted hyperspherical harmonics on S^5, and edges to be dipole transitions $\Delta N = \pm 1$, $\Delta l = \pm 1$ at exact-rational squared matrix elements. **[PANEL-VERIFIED]** At $N_{\max} = 5$ the graph has 56 nodes and 165 edges, all matrix elements computed in exact `fractions.Fraction` arithmetic, with zero irrationals encountered. The single π in the entire Bargmann–Segal ecosystem is the Gaussian normalization π^{-3} of the continuous Bargmann measure, which the discrete graph never invokes.

The HO rigidity theorem

Theorem (Paper 24, HO rigidity). [CONDITIONAL] *The 3D isotropic harmonic oscillator is the unique central potential whose spectrum arises from the Euler number operator on the Hardy space $H^2(S^5)$ restricted to symmetric $SU(3)$ irreps $(N, 0)$.* Paper 24 states the named gap explicitly: the converse half—that SU(3) closure *forces* $V \propto r^2$—is an imported dynamical-symmetry result the paper flags as unverified at the pointer level.

This is the structural dual of the Fock rigidity theorem of Paper 23 (group 4), which states that the S^3 stereographic projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field Hamiltonian onto a function of the Laplace–Beltrami operator on S^3 (its bound states onto the S^3 harmonics) if and only if the spectrum is l-independent within each n-shell, a property unique to $-Z/r$ by $SO(4)$ symmetry. Together the two rigidity theorems (the HO one conditional on the imported $SU(3)$-closure converse) establish that the choice of central potential is sharply tied to the choice of natural manifold: $-Z/r$ to the second-order Riemannian Laplacian on S^3, $\frac{1}{2}m\omega^2 r^2$ to the first-order Euler operator on $H^2(S^5)$.

The Coulomb/HO asymmetry: a seven-layer stratification

Conventions. Sprint and track designations below are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository); "wall" is project shorthand for a structural obstruction established by multiple independent negative results.

[OBSERVATION] A central structural finding of Paper 24, sharpened progressively by the S^5 gauge-structure analysis (Paper 25 [34], QED/gauge group), a sprint on the wedge KMS state of the framework's Lorentzian extension, the spectral-action gravity arc (Paper 51 [39]), and the master Mellin engine, is that the Coulomb and HO discretizations differ along *seven* catalogued layers, identified in stages across 2025–2026, of which five stand: the first was withdrawn on 2026-09-25 and the second on 2026-10-19 (Paper 24). The asymmetry as currently catalogued reads:

1. **Spectrum-computing role of L_0 [withdrawn 2026-09-25].** On the Coulomb S^3 graph, the graph Laplacian L_0 carries the bipartite bound whose saturation $\lambda_{\max} \to 2d_{\max} = 8$ — **[INTERNAL THEOREM]** closed form, at rate $(C_{\text{sat}} + o(1))/n_{\max}^2$ (Paper 0 §VI) — was read as the one convergence indication that survived scrutiny; it is a grid band edge, and L_0 does not converge to Δ_{S^3}. On S^3 as on S^5 the spectrum is a diagonal on the labels (the $SO(4)$ Casimir $n^2 - 1$ there, the Euler operator here), and L_0 is spectrally inert on both, so this layer is not an asymmetry. Its spectrum is bounded in $[0, 8]$ and does *not* approach the continuum $n^2 - 1$ (Paper 7: the spectrum of κL is confined to $[-1/2, 0]$ and no level-by-level Rydberg ladder is reproduced). The matched kinetic scale κ only matches the band edge to the ground state $-\frac{1}{2}$ (an Observation); the hydrogenic levels come from Fock's projection, where the S^3 Coulomb kernel's eigenvalue forces $p_0 = Z/n$ and the energy shell then gives E_n. On the Bargmann–Segal lattice, as on S^3, L_0 encodes transitions only and is not the Hamiltonian (the Wilson reading of L_0 as the matter propagator's inverse does not hold, Paper 24): the spectrum comes from the diagonal Euler operator $\hbar\omega(\hat{N} + 3/2)$, and the graph adjacency encodes dipole transitions only.

2. **Calibration π [withdrawn 2026-10-19].** The Coulomb S^3 projection introduces calibration π, read (a reading, with a **[CONDITIONAL]** engine) as the $k = 0$ volume-slot measure of the master Mellin engine (see Section VII). It is not an asymmetry: both label sets are π-free, and on both sides π enters through a measure and through spectral sums at even argument; in the Kustaanheimo–Stiefel frame the Coulomb projection is linear at fixed energy (Paper 24).

3. **Wilson-physical-content.** On $S^3 = SU(2)$, natural matter coupling for the non-abelian Wilson gauge construction (the gauge field acting on Pauli spinors carried by the graph nodes) is argued, not yet constructed. On S^5, pure-gauge $SU(3)$ Wilson theory on the Bargmann graph is well defined;

what fails is *natural* matter coupling on the $(N,0)$ Hardy tower, because transitions between $(N,0)$ and $(N+1,0)$ irreps are Clebsch–Gordan intertwiners rather than $SU(3)$ group elements, so matter coupling needs a CG intertwiner on every link. The layer is a statement about Wilson gauge with natural matter coupling, not about pure gauge.

4. **Pythagorean orthogonality of the posited wedge generator.** The Coulomb S^3 side admits a hemispheric-wedge KMS state with closed-form Hilbert–Schmidt orthogonality between the posited wedge generator (the diagonal relabelling of Paper 42 Def. 5.1, built from $2J_z$; its modular-Hamiltonian identification is open) and the wedge-restricted Dirac operator. The prefactor $1/\pi^2$ of its closed-form residual is a temporal-compactification π in the sense of Paper 35: it enters through $1/\beta^2 = \kappa_g^2/(4\pi^2)$, the posited normalisation $\beta = 2\pi/\kappa_g$ imported and squared by the Frobenius norm, not Mellin content, and it is not an instance of the master engine's volume slot (the orthogonality itself is a chirality-parity identity independent of β, Paper 32); the HO Bargmann side has no clean analog. This is the operator-algebra-side reading of the asymmetry; for full details see the operator-algebra arc (Group 1).

5. **Spectral-action gravity termination. [SYMBOLIC PROOF]** The Camporesi–Higuchi Dirac spectral action on S^{2m+1} terminates at exactly $(m{+}1)$ power-law terms (Paper 51): only S^3 ($m=1$) gives *pure Einstein gravity* (cosmological + Einstein–Hilbert, no R^2 or higher-curvature corrections), while S^5 ($m=2$) terminates at three terms including an R^2 Gauss–Bonnet contribution. The mechanism is the Bernoulli identity for odd polynomials at half-integer shifts.

6. **Master Mellin engine chirality-parity. [PANEL-VERIFIED]** On the Camporesi–Higuchi S^3 triple at every finite $n_{\max}$, the Mellin source $\mathrm{Tr}(D^k e^{-tD^2})$ exhibits a bit-exact $\mathbb{Z}/2$ chirality-parity factorization (the plain trace vanishes at odd k, the supertrace at even k). On the Bargmann–Segal Hardy sector this fails on both structural conditions: the spectrum is strictly positive (no $\pm$-partner) and no half-integer-weight chirality grading exists on the $(N,0)$ tower—propagating the asymmetry to the master-Mellin level.

7. *CM / Hodge-circle arithmetic of the compact flow.* **[OBSERVATION]** (the flow-level identification is Observation-tier in Paper 56; the S^5 obstructions are exact, Paper 24). On S^3 the compact circle generated by the posited $2J_z$ closure (its identification with a KMS/Bisognano–Wichmann modular flow is **[OPEN]**, Paper 42) has $\pm$-symmetric integer generator spectrum, so its quarter-period point *is* a $\mathbb{Q}$-rational complex structure J, $J^2=-1$, making the fundamental doublet a polarized weight-one $\mathbb{Q}$-Hodge structure with complex multiplication by $\mathbb{Q}(i)$ (Paper 56). The S^5 Hardy phase circle is equally compact and even carries a faithful μ_4 (the metaplectic zero-point $\frac{3}{2}$), yet admits no CM Hodge structure—its generator is strictly positive, the quarter point is $\mathbb{Q}$-rational only where it squares to $+1$, and the conjugate half a weight-one structure needs is exactly what the holomorphic-sector restriction discards. Conductor-four *period* content, by contrast, does transfer (Paper 24's companion non-layer).

The first three layers were catalogued in Paper 24 and Paper 25 (QED/gauge group, S^5 gauge-structure paper); the fourth (wedge KMS) was identified in a 2026 sprint and folded into Paper 24's §V; the fifth (spectral-action gravity) comes from Paper 51; the sixth (master Mellin chirality-parity) from a June 2026 sprint; and the seventh (CM/Hodge-circle arithmetic) from the August 2026 seam arc. Five of the seven stand: the first (spectrum-computing role of L_0) is withdrawn (2026-09-25), because the S^3 graph Laplacian does not converge to the S^3 Laplacian and is spectrally inert, as on S^5, and the second (calibration π) is withdrawn (2026-10-19), because π enters both sides the same ways (Paper 24). The operator-order divide (Paper 31) separates the spectra: the Coulomb side uses a second-order Riemannian Laplacian with a nonlinear stereographic projection (the hydrogen Hamiltonian being the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of it), and the HO side uses a first-order complex-analytic Euler operator with a linear-affine Bargmann projection. The fourth to seventh layers are not placed on that divide by the owner papers (the fifth turns on sphere dimension, the Dirac operator on S^{2m+1}, rather than operator order). The third (Wilson gauge with natural matter coupling) is decided instead by representation theory: on $S^3 = \mathrm{SU}(2)$ matter and links share a representation (argued, not yet constructed), while on S^5 matter coupling needs a Clebsch–Gordan intertwiner.

What the asymmetry says about transcendentals

[OBSERVATION] The bit-exact π-freeness of the Bargmann–Segal lattice matches that of the S^3 label set. Above the labels π enters both sides in the same two ways, through a measure and through spectral sums at even argument, and in the Kustaanheimo–Stiefel frame the Coulomb problem is a constrained four-dimensional oscillator whose lift costs one fibre volume $\pi/4$ (Paper 24). So π is not a structural cost peculiar to the S^3 stereographic projection, and the comparison does not refine the exchange-constant taxonomy of Paper 18 (see Section VII).

VII. THE EXCHANGE CONSTANTS TAXONOMY

[**OBSERVATION**] Paper 18 [25] catalogues where transcendental constants enter physical observables computed from the GeoVac label set. The taxonomy has six exchange-constant tiers plus the algebraic (no-exchange) case—seven categories in all—organized by the type of compactness mismatch each transcendental mediates between the discrete representation and the non-compact coordinate system used for measurement.

Seven categories (as in Paper 18)

1. **Algebraic (no exchange constant).** Content native to the discrete label set and graph: algebraic eigenvalues (the graph's matrix entries are rational), integer multiplicities, exact-rational single-centre Slater F^k integrals (rational arithmetic over $\mathbb{Q}$). π-free in arithmetic over $\mathbb{Q}$.

2. **Intrinsic.** Weyl constants $\pi^{d/2}$ fixed by a manifold's dimension and volume alone, independent of any discretization or basis (e.g. π for $S^1 \to \mathbb{R}$).

3. **Conformal / calibration.** The energy-shell scale $p_0^2 = -2E$ of the stereographic projection and the Fock Jacobian $1/\Omega^4(0) = 1/16$, with which the matched kinetic scale $\kappa = -1/16$ coincides (an Observation, not a derivation; κ only matches the graph's band edge); π entering through the $k = 0$ volume slot of the master Mellin engine, $\mathrm{Vol}(S^2)/4 = \pi$. This is a source of calibration π in the Coulomb sector.

4. **Embedding.** Single transcendental seeds introduced by the interaction of a spectral basis with a coordinate singularity: the seed $e^a \cdot E_1(a)$ that closes the Level-2 prolate-spheroidal radial solver for $m \neq 0$ via partial-fraction decomposition, and the electron-electron cusp $1/r_{12}$. The cusp is two-dimensional in the angular pair coordinate (α, θ_{12}) and cannot be reproduced by any single algebraic insertion; partial-wave Schwartz extrapolation $\sim -A/(l_{\max}+2)^4$ is the standard treatment.

5. **Algebraic-implicit.** Quantities defined by a polynomial equation $P(R, \mu) = 0$ with coefficients in a known field extension, where pointwise diagonalization is a computational convenience rather than a mathematical necessity. The Level-3 adiabatic potential curves $\mu(R)$ on the hyperspherical fiber bundle are algebraic over a polynomial ring whose coefficients lie in $\mathbb{Q}(\pi, \sqrt{2})$.

6. **Composition.** The accuracy cost of factorizing a full N-electron wavefunction into composed coordinates: the factorization destroys inter-group antisymmetry, and the Phillips–Kleinman pseudopotential is its local approximation (Level 5 composed LiH, 144× angular compression at ∼5% R_{eq} error).

7. **Inner-factor input data.** Parameter-tied transcendental content that the framework consumes as Layer-2 input rather than deriving from Layer-1 label data: the Yukawa Dirichlet ring $\mathbb{Q}[y_1^{-2s}, \ldots, y_n^{-2s}]$ generated by the eigenvalues of the finite Dirac operator of the inner factor of a Connes–Marcolli almost-commutative spectral triple. Other external inputs the framework consumes—nuclear-structure scalars ($\langle r^2 \rangle_E$, Zemach radius r_Z, quadrupole moment Q_N) and multi-loop renormalization counterterms—are external input in the same broad sense but are *not* inner-factor data in Paper 18's sense, which is specific to the finite Dirac operator D_F; Paper 36 records the counterterms as external input.

[**OBSERVATION**] The taxonomy sorts the transcendentals observed so far by the projection that introduced them. Its strong form — that the projection type alone determines the transcendental content of any GeoVac observable — is not established: Paper 18 has extended the six tiers with odd-zeta and spinor-intrinsic entries as new cases appeared, and the spinor connection Laplacian's spectral zeta value $\zeta_{\nabla^*\nabla}(2)$ lies outside $\mathbb{Q}[\pi^2]$ (Paper 28), a case the original tiers did not anticipate. Recent work on bound-state QED in the hydrogen-Lamb-shift sprint has confirmed that the framework-native one-loop contributions sit cleanly in the calibration and flow sectors of Paper 34's classification, with the Drake–Swainson structural denominator combinatorial-rational (Paper 36); multi-loop UV-renormalization counterterms remain external input (Paper 36), not inner-factor input data. (Paper 34's "flow" tag has no separate entry in Paper 18's six-tier list; Paper 18 files that content as the infinite-spectral-sum sub-case of the calibration tier, which earlier drafts called "flow".)

The master Mellin engine

[**CONDITIONAL**] A structural sharpening of the taxonomy is the master Mellin engine identified in Paper 18 §III.7 and formalized in the operator-algebra arc of the project. The Weyl–Selberg identification of GeoVac's exchange constants with Mellin transforms of heat kernels

$$\mathcal{M}\left[\mathrm{Tr}\left(D^k \cdot e^{-tD^2}\right)\right](s) \tag{7}$$

indexes three sub-mechanisms by the integer operator order $k \in \{0, 1, 2\}$:

- **M1 ($k = 0$).** Trivial heat kernel collapses to a volume factor $\mathrm{Vol}(S^d)$ (Paper 18: $\mathrm{Vol}(S^2)/4 = \pi$). It is one source of calibration-π content in the Coulomb sector (M2 supplies the π^{2k} of the spectral action);

the Fejér-moment constant $4/\pi$ of Paper 38 shares the value $\mathrm{Vol}(S^2)/\pi^2$ (dual-Coxeter normalisation; $2/\pi = \mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ on the unit S^3) — a reading, not a derivation; a kernel constant, not the convergence rate.

- **M2 ($k = 2$).** Standard Seeley–DeWitt expansion of $\mathrm{Tr}\, e^{-tD^2}$. Its heat-kernel coefficients, via Jacobi θ-inversion on the half-integer-shifted Dirac spectrum, carry Γ-factor powers of $\sqrt{\pi}$; at integer s the spectral-zeta values it produces lie in the pure-Tate even-weight ring $\bigoplus_k \pi^{2k}\mathbb{Q}$ (Paper 55), while at half-integer $s \geq 5/2$ the values are odd zeta values (e.g. $\zeta_{D^2}(5/2) = 14\zeta(3) - \frac{31}{2}\zeta(5)$, Paper 28), which are M3's: a value reached from two slots is placed by the period ring it lands in (Paper 18 §III.7). The heat-kernel and spectral-zeta statements concern different objects and are not in conflict.

- **M3 ($k = 1$).** The plain trace $\mathrm{Tr}(D \cdot e^{-tD^2})$ vanishes on the symmetric Camporesi–Higuchi spectrum; the slot is read through the absolute sum and its vertex-parity grading. This produces the quarter-integer Hurwitz / vertex-parity content (Catalan's G, Dirichlet $\beta(s)$ at integer s) that appears in two-loop QED vertex corrections on S^3. (Slot assignments and ring homes: Paper 18 §III.7, Paper 55 §§3–5, the single owners after the October 2026 sweep.)

The master Mellin engine reading is that M1, M2, M3 are not three independent mechanisms but three subcases of the same Mellin transform on the GeoVac spectral triple, indexed by which power of the Dirac operator appears in the trace. Each sub-mechanism has a characteristic domain of observable: M1 tags volume-slot normalizations (the Fejér-moment constant shares its value, a reading, not a derivation, Paper 38), M2 governs heat-kernel and spectral-action coefficients, and M3 governs vertex-restricted parity-character observables. Mixing the operator order of the question with the operator order of the answer produces the structural degeneracies catalogued in the project's sprint record.

The Layer-1 / Layer-2 split

[**OBSERVATION**] A particularly clean operational consequence of the taxonomy is the two-layer architecture made explicit by Paper 34 (catalogue paper, group 6): the bare combinatorial graph (Layer 1: labels, rational entries, algebraic eigenvalues) versus a finite family of named projections (Layer 2) that add specific variables, dimensions, and transcendentals when the graph is mapped to a physical observable. Paper 34 documents 28 projections to date; each appears in the taxonomy of Paper 18 as either an intrinsic, calibration (whose infinite-spectral-sum sub-case takes Paper 34's "flow" tag), embedding, algebraic-implicit, composition, or inner-factor input data mechanism, or in the algebraic (no-exchange) case. The two papers are complementary: Paper 18 organizes by type of compactness mismatch (what *kind* of transcendental appears), and Paper 34 organizes by named mechanism (*which* projection appears in *which* observable).

VIII. THE UNIVERSAL/COULOMB PARTITION

[**OBSERVATION**] Paper 31 [29] states the cleanest single articulation of what GeoVac claims as universal versus what it claims as potential-tied. The framework is read as a discrete spectral triple $(\mathcal{A}, \mathcal{H}, D)$ in the sense of Connes [12] and the graph-spectral-triple lineage of Marcolli and van Suijlekom [13, 14]: $\mathcal{A}$ is an algebra of functions on the Fock-projected discrete graph, $\mathcal{H}$ is a Hilbert space of scalar and spinor states, and D is a Dirac operator carrying the Camporesi–Higuchi spectrum on S^3 [7]. One caution on which D: the diagonal spectral form D_{GV} (the Camporesi–Higuchi eigenvalues placed on the diagonal of the $|n, \kappa, m_j\rangle$ basis) commutes with $\pi(\mathcal{A})$, which is diagonal in the same basis, so $[D_{\mathrm{GV}}, \pi(f)] = 0$ and that triple carries no metric. The metric, and with it the "Dirac-dependent sector" below, rests on the truthful, non-diagonal Camporesi–Higuchi Dirac operator of Paper 38 [37], not on the diagonal form. (Paper 38's convergence theorem itself, for the commutative factor, is metrized by the translation seminorm; convergence in the Dirac seminorm is open there.)

The A/D split

The structural decomposition reads:

Universal sector ($\mathcal{A}$-only). Results that depend only on the algebra $\mathcal{A}$ and the angular structure of the basis, and *not* on the specific choice of Dirac operator. These transfer free across central potentials. Examples:

- Angular sparsity (Paper 22): ERI density $D(l_{\max})$ depends only on the spherical-harmonic angular factorization.

- Composed scaling (Paper 14, group 4): the exactly Q-linear Pauli-term count of composed-geometry qubit encodings ($N_{\mathrm{Pauli}} = 27.90 \times Q$ for main-group molecules at fixed basis) depends on the composed block decomposition and, within blocks, the Gaunt selection rules.

- Selection rules: $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ for dipole transitions; the $3j$-triangle inequality at every vertex of the two-body integral.

Potential-specific sector (D-dependent). Results that depend on the specific spectrum and matrix ele-

ments of the Dirac operator D, and therefore on the specific central potential. These require per-system rederivation. Examples:

- The spectrum itself: $n^2 - 1$ for Coulomb, $N + 3/2$ for HO.
- The matched scale $\kappa = -1/16$ (Coulomb-specific; no analog on the Bargmann lattice; it matches only the graph's band edge, the energies entering through the energy shell).
- Transcendental content of spectral invariants: the Coulomb spectral zeta function lives in the π-polynomial ring; other systems may live in different rings (the HO Euler ladder at $s = 4$ also carries $\zeta(3)$, Paper 24).
- Rigidity theorems: Fock rigidity (Paper 23) pins the Coulomb geometry to S^3; HO rigidity (Paper 24; conditional on the imported $SU(3)$-closure converse) pins the HO geometry to $H^2(S^5)$ with $(N, 0)$ symmetric irreps.
- The candidate fine-structure-constant combination $K = \pi(B+F-\Delta)$ of Paper 2 (an Observation): lives in the Coulomb-specific sector through the volume-slot π on S^3 (Paper 31); of its three ingredients only Δ uses the Dirac degeneracy.

Why the partition matters operationally

The partition organizes Papers 22, 23, 24, and 25 into a single conceptual structure. Paper 22's angular sparsity theorem lives in the universal sector and transfers free between hydrogen and the deuteron; Paper 23's nuclear shell model qubit Hamiltonians inherit the angular sparsity automatically but require a fresh per-potential analysis of the Dirac sector. Paper 24's HO rigidity lives in the potential-specific sector and explicitly does *not* transfer to the Coulomb side—the Bargmann-style first-order operator has no analog on S^3. Paper 25 (group 5, QED/gauge) on the SU(2) Wilson lattice gauge structure of the Hopf graph also lives in the potential-specific sector as far as Wilson gauge *with natural matter coupling* is concerned: that depends on the parallelizability of $S^3 = SU(2)$, which has no analog on S^5 (pure-gauge $SU(3)$ is well defined on S^5; natural $(N, 0)$ matter there needs a Clebsch–Gordan intertwiner).

The operational utility of the partition is that anyone porting GeoVac to a new system can read off, from the universal sector, what structural properties they inherit for free, and from the potential-specific sector, what calibrations they need to derive afresh. For nuclear shell models, gauge-theoretic extensions, and oscillator-based quantum dots, the universal sector provides angular sparsity and composed scaling automatically; the potential-specific sector has to be redone for each system.

Status as a spectral-triple-language reformulation

Paper 31 is careful to frame the partition as a structural reformulation, not a new theorem. The angular sparsity theorem (Paper 22), the Fock rigidity theorem (Paper 23), and the HO rigidity theorem (Paper 24; conditional on the imported $SU(3)$-closure converse) are each established results; the partition is the observation that they decompose cleanly along the algebra-only / Dirac-dependent divide. The framing uses the language of NCG and spectral triples because that language has the structural vocabulary to articulate the divide; the underlying physics content is unchanged.

IX. THE RECONVERGENCE: TWO-BODY STRUCTURE, PERIODS, AND THE TANNAKIAN SUBSTRATE

The foundations arc described so far *fans out* from a single taproot. The packing construction (Paper 0), made physical by the Fock S^3 projection (Paper 7), is the root; from it the framework branches into spectral graph methods (Paper 1), the exchange-constant taxonomy (Paper 18), angular sparsity (Paper 22), the Bargmann–Segal lattice (Paper 24), and the $\mathcal{A}/D$ partition (Paper 31). Four more recent papers do the opposite—they *reconverge*, looping the scattered structural findings back into a single foundational statement about what the framework autonomously generates and what it consumes.

A. Two-body structure (Paper 54)

The $\mathcal{A}/D$ partition of Paper 31 is, in the single-particle setting, a sorting of results into algebra-only and Dirac-dependent. Paper 54 tests it at the two-body level by constructing the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$ and asking whether the gauged spectral action generates the inter-particle Coulomb interaction from the framework's own structure. The free tensor-product Dirac $D_{\text{total}} = D_1 \otimes I + \gamma_1 \otimes D_2$ satisfies $\{D, \gamma\} = 0$ exactly, so the free heat trace factorizes and the free spectral action has no cross term and contains no interaction—the correct statement that free particles do not interact. **[MEASURED]** Gauging via the inner fluctuation of the truncated operator-system algebra produces a genuine, non-factorizable two-body interaction (connected fraction 77% at $n_{\max} = 2$, falling to 32% at $n_{\max} = 3$— cutoff-sensitive, and measured at only two cutoffs). **[MEASURED]** It is supported only on the direct (density–density) elements, so it is m-conserving, $\Delta l = 0$ and pure $k = 0$ by construction: a strict subset of the Coulomb channels, with no exchange or $k > 0$ term. Its radial weights, however, are *not* proportional to the Coulomb $1/r_{12}$ kernel (Pearson $r = 0.41$–0.58, *decreasing* with $n_{\max}$; Paper 54 reads this as structural

rather than finite-size, an inference from the same two cutoffs). This delineates the partition at the two-body level: the angular algebra determines *which* channels may couple (universal across central potentials by Paper 22), and the Dirac operator / metric determines *how strongly* (potential-specific). The recurring "multi-focal composition wall" sits on this $\mathcal{A}/D$ boundary; the heat-kernel versus Green's-function contrast does not explain it, since the resolvent route fails the same way (Papers 31, 54), and the mismatch is attributed to the Fock projection's conformal factor.

B. Periods (Paper 55)

The master Mellin engine of Paper 18 indexes the framework's π-content by operator order $k \in \{0, 1, 2\}$ in $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$. Paper 55 extends this engine into a period-theoretic census. **[INTERNAL THEOREM]** The ring memberships of the census values below are theorems; their indexing by Mellin slot is a reading, and the underlying case-exhaustion is **[CONDITIONAL]**; the scope is a census (**[OBSERVATION]**) of S^3 spectral-zeta and heat-trace data—it does not cover the chemistry integrals (e.g. the embedding seed $e^a E_1(a)$), spectral determinants such as $\pi\, e^{\zeta(3)/(2\pi^2)}$, or the genus-one three-centre period of Paper 61. That census comprises integer powers of π, odd zeta values $\zeta(2k{+}1)$, Catalan $G = \beta(2)$ and Dirichlet-β values, and the depth-k loop tower as identified so far ($k \le 3$), all of which sit in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathbb{Z}[i, 1/2]$, read by Mellin slot: $M_1 \subset \mathbb{Q}[\pi, \pi^{-1}]$ (pure-Tate, depth 0), $M_2 \subset \bigoplus_k \pi^{2k}\mathbb{Q}$ at integer s (pure-Tate even-weight, depth 0, a strict refinement of the Fathizadeh–Marcolli classification [47] specialized to the static S^3 sub-case), and $M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4 (cyclotomic mixed-Tate, via Hurwitz zeta at quarter-integer shifts and the Deligne–Glanois descent [48]; the vertex-parity-as-χ_{-4} correspondence of Paper 28 [35]). This is the period-theoretic extension of the $\mathcal{A}/D$ partition: the Coulomb-(D-)sector spectral transcendentals are pinned to named cyclotomic mixed-Tate rings rather than chased indefinitely. (Discipline check folded in here: the constant $S_{\min}$, long carried as "PSLQ-irreducible," is now resolved as $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ in closed form—the prior label was a basis-coverage artifact.)

C. The Tannakian substrate and the cosmic-Galois comparison homomorphism (Paper 56)

If the framework's periods all live in a single named period ring, the natural categorical home is a motivic Galois group. **[PANEL-VERIFIED]** Paper 56 assembles, bit-exact on rational panels (thousands of zero residuals across $n_{\max} \in \{2, 3, 4\}$), the Tannakian substrate of the GeoVac spectral triple at finite cutoff: a cofiltered pro-system of finite-cutoff Hopf-algebra truncations, an abelian symmetric-monoidal rigid representation category, a faithful exact $\otimes$-fiber functor, and an explicit inclusion of the candidate cosmic-Galois group $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N} \rtimes SL_2$ into $\mathrm{Aut}^{\otimes}(\omega)$. The cosmic-Galois comparison is **[INTERNAL THEOREM]** at depth 1 only: sending each M3 generator to the depth-1 de Rham class of its level-4 motivic period, and each M1, M2 generator to 0, gives a homomorphism $\mathcal{U}_4 \to \mathbb{G}_a^{3N}$ *from* the unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ (Deligne [8], Glanois [48]) *into* GeoVac's abelian factor, factoring through $\mathcal{U}_4^{\mathrm{ab}}$, with image of dimension at most 2 per weight. Paper 56's earlier statement, a homomorphism $U^*_{\mathrm{GV}} \to \mathcal{G}_4$, had the arrow reversed and is withdrawn (2026-11-03); only its label and the symbol Φ^{inj} keep the older "injection" name. The comparison group is the *level-4* group, not Brown's $\mathcal{G}_{\mathrm{MT}(\mathbb{Z})}$: Eskandari–Murty–Nemoto [9] construct a motive with period G that is mixed Tate over $\mathbb{Q}(i)$ but not over $\mathbb{Q}$; the level-4 placement is the classical $G = \mathrm{Im}\,\mathrm{Li}_2(i)$ in Deligne's $N = 4$ setting. That G is *not* a period of $\mathrm{MT}(\mathbb{Z})$ is expected but not proven, so this motivates the refinement rather than forcing it. Two scope qualifiers are essential: (i) the comparison sees only $\mathcal{U}_4^{\mathrm{ab}}$ (Reading A)—GeoVac's primitive Hopf substrate is cocommutative by construction; at the JLO entire-cyclic-cocycle level at depth 2 only the even flavour is S_3-symmetric, so the probe does not decide the reading—and M1, M2 drop out; (ii) depth-≥ 2 level-4 content, which a primitive substrate cannot receive, is the named deep forward question.

The elliptic layer (Papers 59 and 61 [44, 45]). The comparison above is with the *mixed-Tate* (genus-zero) cosmic Galois; the elliptic (genus-one) layer that Paper 56 reaches for—a Hain–Brown mixed-elliptic-motive target [49, 50]—came back negative on the abstract Sym^2 panel, structurally because GeoVac's periods are $\mathbb{Q}(i)$-CM-pure rather than generically modular. **[MEASURED]** Paper 59 supplies a concrete elliptic period from the chemistry side: the radial core of the three-center electron-repulsion integral, read in momentum space, is a two-scale Bessel moment on the Legendre / $\Gamma(2)$ family; the modular and cosmic-Galois structure that follows—the family arithmetic, the $X_0(2)$ descent, the 66-digit certification and the guarded searches—is developed in the companion Paper 61 [45]. The load-bearing arithmetic is *fibre-independent*: the fibre period is $K = \frac{\pi}{2}\theta_3^2$ and θ_3^2 is the theta series of $\mathbb{Z}[i]$, so the family carries conductor-4 arithmetic at every fibre—the same $\mathbb{Q}(i)$ the Dirac Kramers doublet of Paper 56 realizes, by a *different* forcing (level rather than spin): the two routes are convergent, not common-cause, and no comparison map between them has been exhibited. The physical density exchange acts on the modulus as $\rho \mapsto 1/\rho$, so the observable descends to $X_0(2) = X(2)/\langle\iota\rangle$; its physically distinguished fibre is the coincident-scale *pure-Tate cusp* $\rho = 1$, while the $\tau = i$ CM fibre is a visited member of the exchange orbit $\{\frac{1}{2}, 2\}$, not a pinned one.

The radial core is an Eisenstein / complex-multiplication *Bessel moment*—an irregular (exponential) period, of which a $\Gamma(2)$ multiple modular value is only the regular $D \to 0$ shadow; the integrated observable is *not* a cusp-form L-value, and as a function of the bond angle it leaves the two-center class (Paper 59). Its value is now certified to 66 digits, and the guarded searches exclude clean forms: no low-height classical combination (through weight three, the discriminant-8 ring through weight two), including the Catalan-completed ring at height ≤ 10—so no *clean* conductor-4 form of the physical value exists at those heights (the arithmetic lives in the family and the shadow), and its explicit finite realisation remains open. Two honest bounds keep this in proportion. First, the level is $\Gamma(2)$, the classical *universal* Legendre level (below the two-loop sunrise's $\Gamma_1(6)$): landing on the Legendre family is expected of any four-branch-point curve, so the informative content is the rational modulus map $\lambda = 1 - \rho$, the $X_0(2)$ descent, and the physical Bessel twist's level promotion $\Gamma(2) \to \Gamma(4)$, not an exotic level. Second, whether the observable reaches the *generic* (non-CM, full-SL_2, $E_4/E_6/\Delta$) Hain–Brown locus is open. **[SYMBOLIC + MEASURED]** What the irreducibility *does* certify is structural: because the rank-four connection is the Fourier–Laplace transform of an irreducible rank-one connection—hence simple, with no invariant *coordinate* subspace in its monodromy frame—no closed form arises through any factorization, so the period is not reducible to the mixed-Tate layer by factorization (the genus-one label itself is set by the curve's two distinct Fock scales). On the physical contour the two layers demonstrably touch at the pure-Tate diagonal cusp; the seam's two upgrade tests—a natural comparison map, and a measured $\beta(2)$ in the physical value—are respectively tautology-blocked and answered negative at clean heights. The elliptic layer is thus a concrete, irreducible genus-one *instance* that corroborates rather than lifts the Hain–Brown negative—not filled to the generic modular locus.

D. The forced/free seam (Paper 57)

[OBSERVATION] The reconvergence closes with an observations paper that consolidates, across the corpus, the boundary between what the framework forces and what it consumes. Paper 57 catalogues the forced entries (forced relative to its axiom set, which includes the adopted Upgrade B axiom; the label-set spectrum n^2-1, carried on the labels as the $SO(4)$ Casimir—not the graph Laplacian's grid band edge, which Paper 57 records as matched rather than forced—the S^3 outer manifold, the gauge group $U(1) \times SU(2) \times SU(3)$ from Bertrand $\times$ Hopf-tower truncation (an organizing reading plus an adopted axiom, not a derivation), the master Mellin engine, two-term-exact gravity), the single admitted-not-forced entry (the quaternionic factor $\mathbb{H}$), and the calibration entries (Yukawa values, generation count, cut-off moments, atomic screening exponents, multi-loop QED counterterms). It identifies the cross-domain unity pattern—the same multi-focal composition wall recurs in seven settings, the two nuclear ones sharing the textbook matching mechanism of nonrelativistic QED—and tests a packing-reachability discriminator (an entry is forced iff a witness derivation exists from the packing axiom, the real-spectral-triple axioms, Hopf-tower truncation, Bertrand, Upgrade B and the CCM canonical Standard-Model representation). The packing-reachable tag is the F/C label by construction, so its 98.3% agreement across the 60-entry catalogue of June 2026 is an internal-consistency check, not an independent discriminator (Paper 57 §6.1); the genuine discriminator signal is the two-family decomposition below. (The catalogue's released-axis sub-domain J, added in October 2026, and its entry counts are kept in `docs/forced_free_seam.md`; the door-indexed view is `docs/atlas.md`.) The free side splits into two structurally distinct families: a multi-focal composition family (the framework can build it but cannot autonomously pick a number) and an inner-factor input-data family (sharing no generator with the outer-factor rings, an Observation-level reading). This is the foundational reconvergence: the period census's theorem-grade ring memberships (Paper 55) and the finite-cutoff Tannakian comparison homomorphism (Paper 56) bound the empirical forced/free observation (the census's scope is an **[OBSERVATION]**)—with the $\mathcal{A}/D$ partition (Paper 31) supplying the spectral-triple language rather than a further theorem— though Paper 57 holds the forcing-iff-property principle itself open as a preliminary observation (it is an observations paper, not a theorem paper). Read this way, the framework maps the compact, Peter–Weyl, discrete skeleton of the systems it treats and consumes, rather than generates, the continuum calibration data the skeleton does not reach.

X. OPEN QUESTIONS AND OUTLOOK

The foundations arc as catalogued here leaves five named structural questions open.

The α combination rule

Paper 2 [22] (group 5, QED/gauge) presents the candidate combination

$$K = \pi(B + F - \Delta), \qquad K \stackrel{?}{=} \alpha^{-1}, \tag{8}$$

where $B = 42$ is a finite Casimir trace at the truncation $n_{\max} = 3$, $F = \zeta_R(2) = \pi^2/6$ is the Fock-degeneracy Dirichlet series at the packing exponent $d_{\max} = 4$, and $\Delta = 1/40$ is the inverse Dirac mode count $g_3^{\mathrm{Dirac}}(S^3) = 2(n+1)(n+2)|_{n=3}$. Each ingredient now has a structurally identified spectral home; twelve mechanisms have been eliminated as candidate single-principle derivations

of the sum [22]. The combination rule itself remains an *Observation* at the rule level (never a conjecture or derivation, per the project's hard rule): there is no first-principles derivation explaining why the additive combination of three objects with spectral homes of distinct type approximately equals α^{-1} (the plain sum K matches α^{-1} to 4.8×10^{-7} relative; the 8.8×10^{-8} agreement is that of Paper 2's self-consistency root $\alpha^{-1} + \alpha^2 = K$). The synthesis treats this as an open structural question and does not assert priority for the observed identification.

The second packing axiom (W3)

[OPEN] The packing construction of Paper 0 reproduces the per-shell count of the angular sector (l, m) and provides natural correspondences for n and spin multiplicity, but it does not produce calibration data such as Yukawa couplings or the number of fermion generations. These appear in the exchange-constant taxonomy as Layer-2 inner-factor input data: the framework consumes them rather than deriving them. The open question labeled W3 in the project's internal taxonomy is whether a second packing-axiom-like construction generates calibration data from a similarly minimal principle. A 2026 sprint surveying candidates in the spectral-zeta / master-Mellin-engine ring returned uniformly negative results across PMNS angles, lepton masses, and a $\sim$ 21,000-form mechanical basis [30]. The second-packing-axiom question is open and is structurally distinct from "find a packing on a different geometry"—the latter has been answered for the HO via the Bargmann–Segal lattice without addressing W3.

Cross-manifold extensions

The Coulomb and HO discretizations live on S^3 and the Hardy sector of S^5 respectively. **[OPEN]** A natural question is whether a coupled system (electronic plus nuclear, or two distinct potential regimes within one molecule) admits a tensor-product spectral triple of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$. Within the foundations arc, Paper 24's Coulomb/HO asymmetry (five standing layers) documents structural obstructions to such a construction at the NCG-framework level. The operator-algebra arc (Group 1) has formally named this as a frontier-of-field open problem; we are not aware of a construction that supports this cross-manifold tensor product.

The multi-focal composition wall

[OPEN] The framework's composition rules for multiple Fock-style projections constitute an active research direction. The internal taxonomy labels three sub-walls: W1 (within-framework sub-walls: cross-register coordinate operators, magnetization-distribution operators, frozen-core cross-center screening), W2 (multi-loop UV/IR composition and cross-manifold extensions), and W3 (inner-factor calibration data, discussed above). W1c has been built without closing the chemistry wall it targeted (the W1d/W1e residual, Paper 32); W1a/b are architectural gaps; W2a (multi-loop renormalization) remains open at the field-theory frontier of the operator-algebra arc. None of these walls falsify the foundations as presented in this synthesis; they sharpen the open-question boundary of the framework.

The elliptic layer of the cosmic Galois

[OPEN] Paper 56's cosmic-Galois comparison is with the mixed-Tate (genus-zero) group $\mathcal{G}_4$; the elliptic (genus-one) layer it reaches for—the Hain–Brown mixed-elliptic-motive target over $M_{1,1}$—came back negative, because GeoVac's periods are $\mathbb{Q}(i)$-CM-pure rather than generically modular. Paper 59 now instantiates that layer with a concrete, irreducible genus-one period (the three-center integral's elliptic Bessel moment on the Legendre/$\Gamma(2)$ family, descending to $X_0(2)$), carrying the same $\mathbb{Q}(i)$ arithmetic at the *family* level (θ_3^2, the $\mathbb{Z}[i]$ theta series) by level-forcing rather than spin-forcing—convergent with the Dirac route, not common-cause. The open question is whether any GeoVac observable reaches the *generic* (non-CM, full-SL_2) Hain–Brown locus—a lift that would unify the genus-zero and genus-one layers (the Hain–Brown mixed-elliptic completion). On the physical contour the two layers demonstrably meet at the pure-Tate cusp; no natural comparison map has been exhibited, and the measured seam test ($\beta(2)$ in the physical value) is negative at clean heights—a lift beyond convergence is unestablished.

XI. CONCLUSION

The Group 3 foundations arc of GeoVac establishes a small set of structural results that together provide a coherent discretization of non-relativistic quantum mechanics on natural geometries. **[SYMBOLIC PROOF]** The conformal chain from the unit-S^3 Laplace–Beltrami operator onward is audited symbolically (Paper 7 [24]: Fock's projection makes the bound-state Hamiltonian $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$, its non-definitional kernel-eigenvalue step tested exactly for $n \leq 4$, general n Fock's); **[WALL]** the discrete Fock graph's edge Laplacian does *not* converge to that operator (it splits into $n_{\max}$ disconnected l-blocks with an $n_{\max}$-dimensional kernel), while the label set is exact, carrying $-\Delta_{S^3}$ as the diagonal $SO(4)$ Casimir $n^2 - 1$; and **[INTERNAL THEOREM]** the state-space GH convergence of the spinor spectral truncations is proved in Paper 38 [37] for the commutative S^3 factor, on the true Camporesi–Higuchi window (the almost-commutative extension is **[OPEN]**): the

bound and $\gamma_{n_{\max}} \to 0$ are proved (Paper 38, main theorem: the Fejér moment is $(4/\pi)\log n_{\max}/n_{\max}$, **[INTERNAL THEOREM + MEASURED]**; a tapered amplitude gives a **[MEASURED]** $O(1/n_{\max})$); **[PRIOR ART mechanism; explicit densities new]** the angular sparsity of two-body matrix elements is potential-independent (Paper 22 [27]); the 3D harmonic oscillator admits a bit-exactly π-free discrete encoding on $H^2(S^5)$ (Paper 24 [28]); **[OBSERVATION]** the framework's results decompose cleanly into an algebra-only universal sector and a Dirac-dependent potential-specific sector (Paper 31 [29]); the exchange-constant taxonomy of Paper 18 [25] catalogues where transcendentals enter physical observables and identifies a **[CONDITIONAL]** master Mellin engine at three operator orders as the generator of the π-content of the projection chains case-checked so far (Paper 32 [36]'s case-exhaustion theorem, over Paper 34 §III.1–15, with its (i)⇔(ii) leg resting on empirical verification); the packing axiom of Paper 0 [19] reproduces the per-shell count of the angular sector geometrically (its structural choices motivated, not forced); the spectral graph theory of Paper 1 [20] and the sparse $O(V)$ time propagation of Paper 6 [23] (archived; demonstrated on the graph model, not physical hydrogen) provide the computational machinery.

The framework's concrete advantages are independent of the interpretive question of which description—discrete or continuous—is more fundamental. $O(V)$ sparsity, angular momentum selection rules baked into the basis, bit-exactly rational squared matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound at fixed $l_{\max}$ are statements about the discrete label set and the basis it indexes, not about the graph Laplacian (the integer spectrum n^2-1 belongs to the continuum operator, carried exactly on the labels as the diagonal $SO(4)$ Casimir, while the graph Laplacian's own spectrum is closed form, bounded in $[0,8]$, and does not converge to it — Paper 0 §VI, Paper 7); their continuum counterparts are theorems about the Laplace–Beltrami operator or the Euler operator on the corresponding natural manifold, and consistency between the two is guaranteed by the label-set/Casimir correspondence together with Fock's conformal equivalence. The discrete label lattice and the continuous operator are dual descriptions linked by an exact Peter–Weyl label identification, with Fock's conformal projection linking the operator to the Schrödinger equation ($H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$), and the interpretive question is left open.

The open questions catalogued in Section X are substantive but do not destabilize the structural results of the foundations arc. The α combination rule remains an Observation; the second-packing-axiom question is open; no construction for the cross-manifold extensions is known at the NCG-framework level; the multi-focal composition wall is an active research frontier; and the elliptic (genus-one) layer of the cosmic Galois is instantiated but not lifted—the generic Hain–Brown locus stays open. Of the composition walls, W2a and W2b sit at the frontier of the spectral-action program, while W1a–W1c are GeoVac-internal (Paper 32 [36]); none is a foundational concern.

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[1] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).
[2] V. Bargmann, "Zur Theorie des Wasserstoffatoms: Bemerkungen zur gleichnamigen Arbeit von V. Fock," Z. Phys. **99**, 576 (1936).
[3] M. Bander and C. Itzykson, "Group Theory and the Hydrogen Atom," Rev. Mod. Phys. **38**, 330 and 346 (1966).
[4] A. O. Barut and H. Kleinert, "Transition Probabilities of the Hydrogen Atom from Noncompact Dynamical Groups," Phys. Rev. **156**, 1541 (1967).
[5] V. Bargmann, "On a Hilbert space of analytic functions and an associated integral transform, Part I," Commun. Pure Appl. Math. **14**, 187 (1961).
[6] B. C. Hall, "The Segal–Bargmann coherent state transform for compact Lie groups," J. Funct. Anal. **122**, 103 (1994).
[7] R. Camporesi and A. Higuchi, "On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces," J. Geom. Phys. **20**, 1 (1996).
[8] P. Deligne, *Le groupe fondamental unipotent motivique de* $\mathbb{G}_m - \mu_N$*, pour* $N = 2, 3, 4, 6$ *ou* 8, Publ. Math. IHÉS **112** (2010) 101–141.
[9] P. Eskandari, V. K. Murty, and Y. Nemoto, *Mixed motives and linear forms in the Catalan constant*, arXiv:2510.20648 (2025).
[10] J. Avery, *Hyperspherical Harmonics: Applications in Quantum Theory* (Kluwer Academic, Dordrecht, 1989).
[11] V. Aquilanti and A. Caligiana, "Sturmian approach to one-electron many-center systems: integrals and iteration schemes," Chem. Phys. Lett. **366**, 157 (2002).
[12] A. Connes, *Noncommutative Geometry* (Academic Press, San Diego, 1994).
[13] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," J. Geom. Phys. **75**, 71 (2014); arXiv:1301.3480.
[14] C. I. Pérez-Sánchez, "Bratteli networks and the Spectral Action on quivers," arXiv:2401.03705 (2024).

[15] E. U. Condon and G. H. Shortley, *The Theory of Atomic Spectra* (Cambridge University Press, Cambridge, 1935).

[16] J. C. Slater, *Quantum Theory of Atomic Structure*, Vols. I and II (McGraw-Hill, New York, 1960).

[17] J. Suhonen, *From Nucleons to Nucleus: Concepts of Microscopic Nuclear Theory* (Springer, Berlin, 2007).

[18] L. C. Biedenharn and J. D. Louck, *Angular Momentum in Quantum Physics* (Addison-Wesley, Reading, MA, 1981).

[19] J. Loutey, "Counting Angular Momentum Labels by Isotropic Packing," GeoVac Paper 0 (2026).

[20] J. Loutey, "The Geometric Atom: the Hydrogen State Space as a Label Lattice," GeoVac Paper 1 (2026).

[21] V. G. Kac, *Infinite Dimensional Lie Algebras*, 3rd ed. (Cambridge University Press, Cambridge, 1990).

[22] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," GeoVac Paper 2 (group 5, 2026).

[23] J. Loutey, "$O(N)$ Quantum Dynamics and Molecular Spectroscopy via Spectral Graph Theory," GeoVac Paper 6 (2026). [Archived: superseded within the series; retained for provenance.]

[24] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[25] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[26] J. Loutey, "The Geometric Vacuum: Quantum Structure from Compact Manifold Topology," GeoVac Paper 21 (Synthesis, 2026).

[27] J. Loutey, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," GeoVac Paper 22 (2026).

[28] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026).

[29] J. Loutey, "The Universal/Coulomb Partition of the GeoVac Spectral Triple," GeoVac Paper 31 (2026).

[30] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (Observations, 2026).

[31] I. R. Klebanov, S. S. Pufu, B. R. Safdi, "F-Theorem without Supersymmetry," *Journal of High Energy Physics* **10** (2011) 038, arXiv:1105.4598.

[32] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," GeoVac Paper 14 (2026).

[33] J. Loutey, "The Periodic Table Recast as S_N Representation Theory on S^{3N-1}," GeoVac Paper 16 (2026).

[34] J. Loutey, "The Hopf Graph as a Lattice Gauge Structure: A Framework Observation," GeoVac Paper 25 (2026).

[35] J. Loutey, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," GeoVac Paper 28 (2026).

[36] J. Loutey, "The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage," GeoVac Paper 32 (2026).

[37] J. Loutey, "State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3," GeoVac Paper 38 (2026).

[38] J. Loutey, "State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric," GeoVac Paper 40 (2026).

[39] J. Loutey, "Gravity from the GeoVac spectral action: continuum results and the discrete-substrate program," GeoVac Paper 51 (2026).

[40] J. Loutey, "Two-body interactions from the tensor-product spectral triple: Selection rules from the algebra, coupling strengths from the metric," GeoVac Paper 54 (2026).

[41] J. Loutey, "Periods of GeoVac: a cyclotomic mixed-Tate census of the master Mellin engine," GeoVac Paper 55 (2026).

[42] J. Loutey, "The Tannakian substrate of GeoVac," GeoVac Paper 56 (2026).

[43] J. Loutey, "Forced and Free Content in the GeoVac Framework: An Observations Paper," GeoVac Paper 57 (2026).

[44] J. Loutey, "The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: A Genus-One Radial Core at the Third Center in Momentum-Space Slater Theory," GeoVac Paper 59 (2026).

[45] J. Loutey, "The Bessel-Moment Period Algebra of the Three-Center Coulomb Integral: Modular Structure and Cosmic-Galois Placement," GeoVac Paper 61 (2026).

[46] Y. Gaudillot-Estrada and W. D. van Suijlekom, "Convergence of spectral truncations for compact metric groups," Int. Math. Res. Not. IMRN **2025**, no. 13, rnaf197 (2025); arXiv:2310.14733.

[47] F. Fathizadeh and M. Marcolli, "Periods and motives in the spectral action of Robertson–Walker spacetimes," Comm. Math. Phys. **356**, 641–671 (2017); arXiv:1611.01815.

[48] C. Glanois, "Motivic unipotent fundamental groupoid of $\mathbb{G}_m \setminus \mu_N$ for $N = 2, 3, 4, 6, 8$ and Galois descents," J. Number Theory **160**, 334–384 (2016); arXiv:1411.4947.

[49] R. Hain, "The Hodge-de Rham theory of modular groups," in *Recent advances in Hodge theory* (LMS Lecture Notes 427, Cambridge Univ. Press, 2016) 422–514; arXiv:1403.6443.

[50] F. Brown, "Multiple modular values and the relative completion of the fundamental group of $M_{1,1}$," arXiv:1407.5167 (2014; v4, 2017).